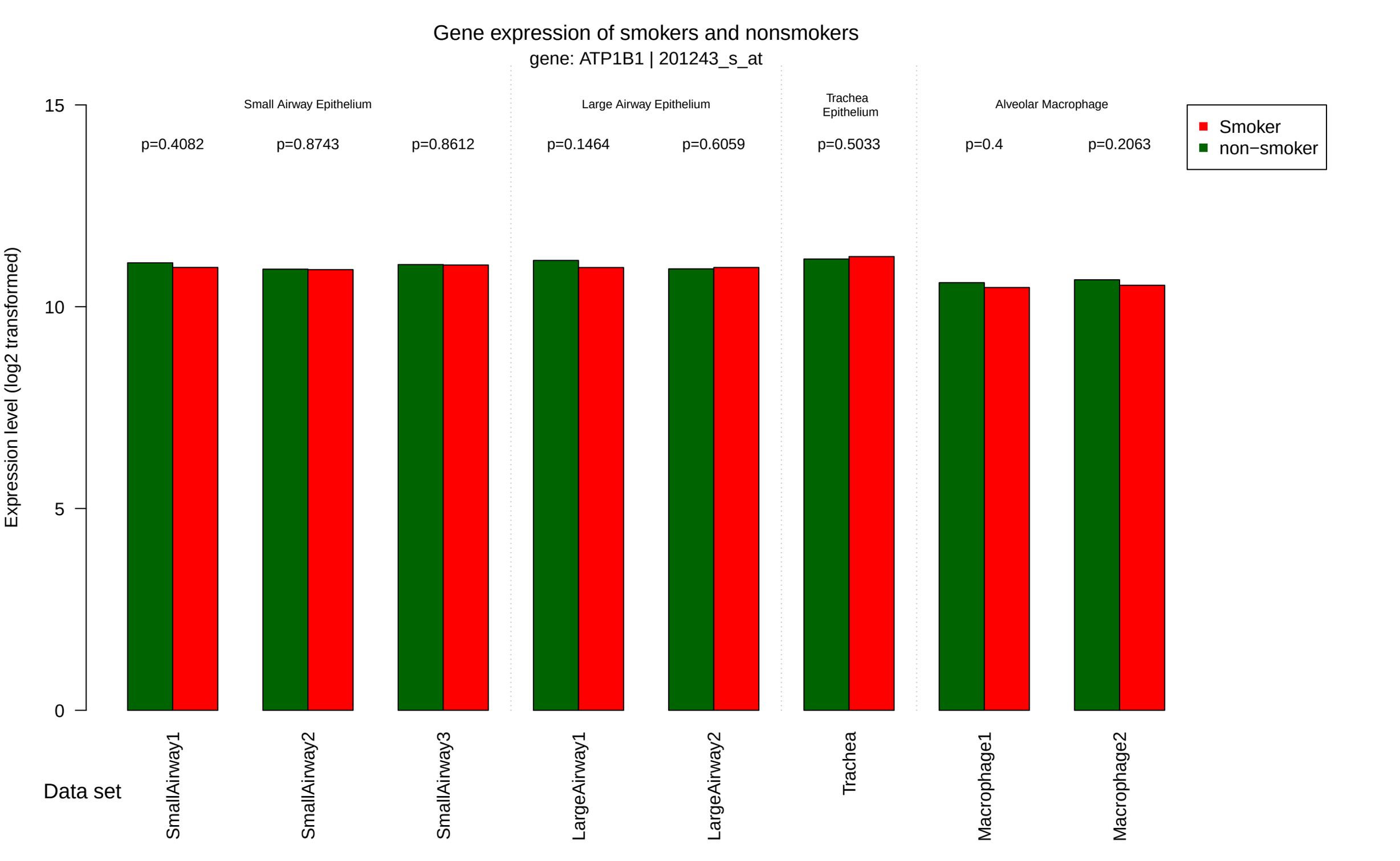
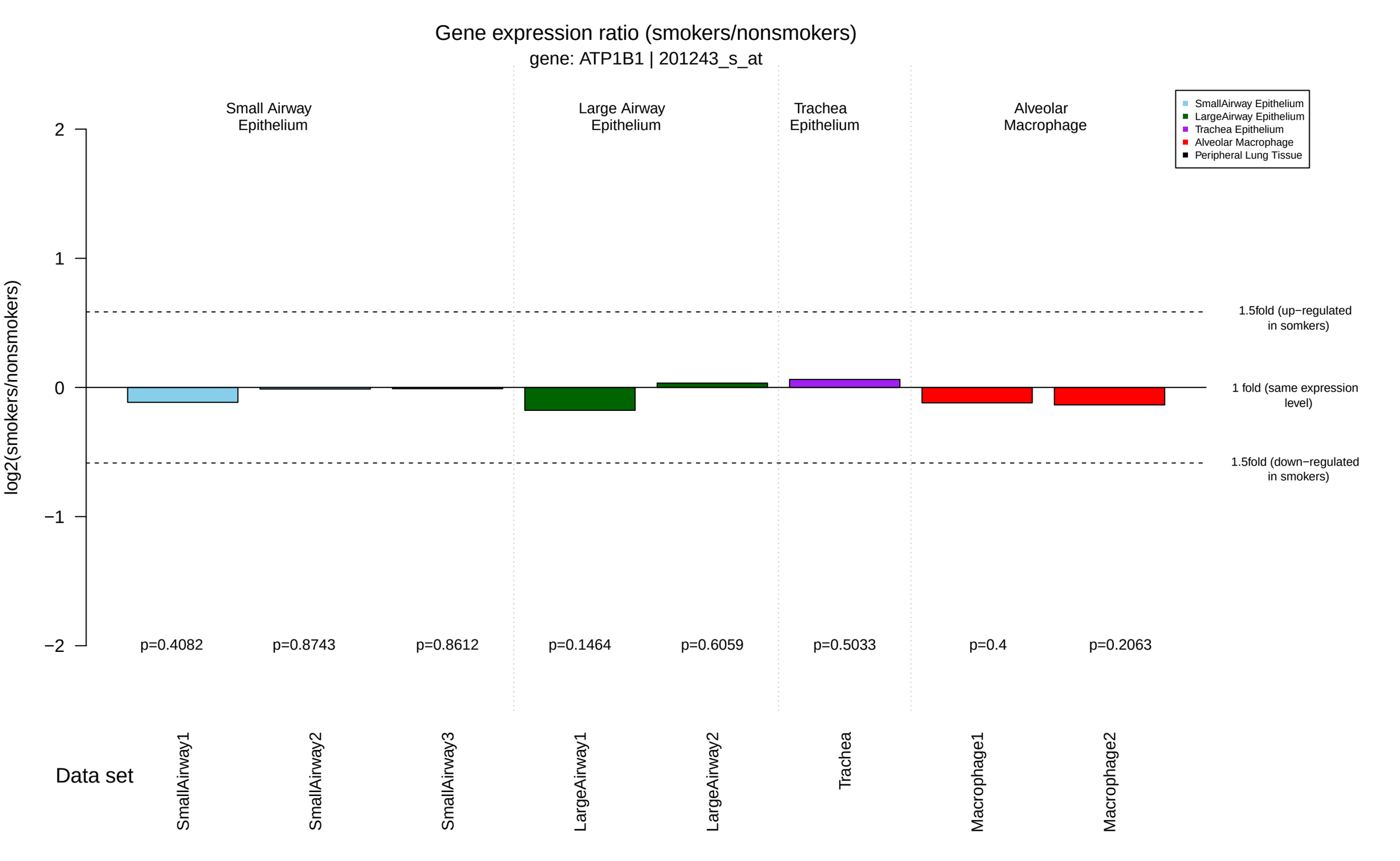
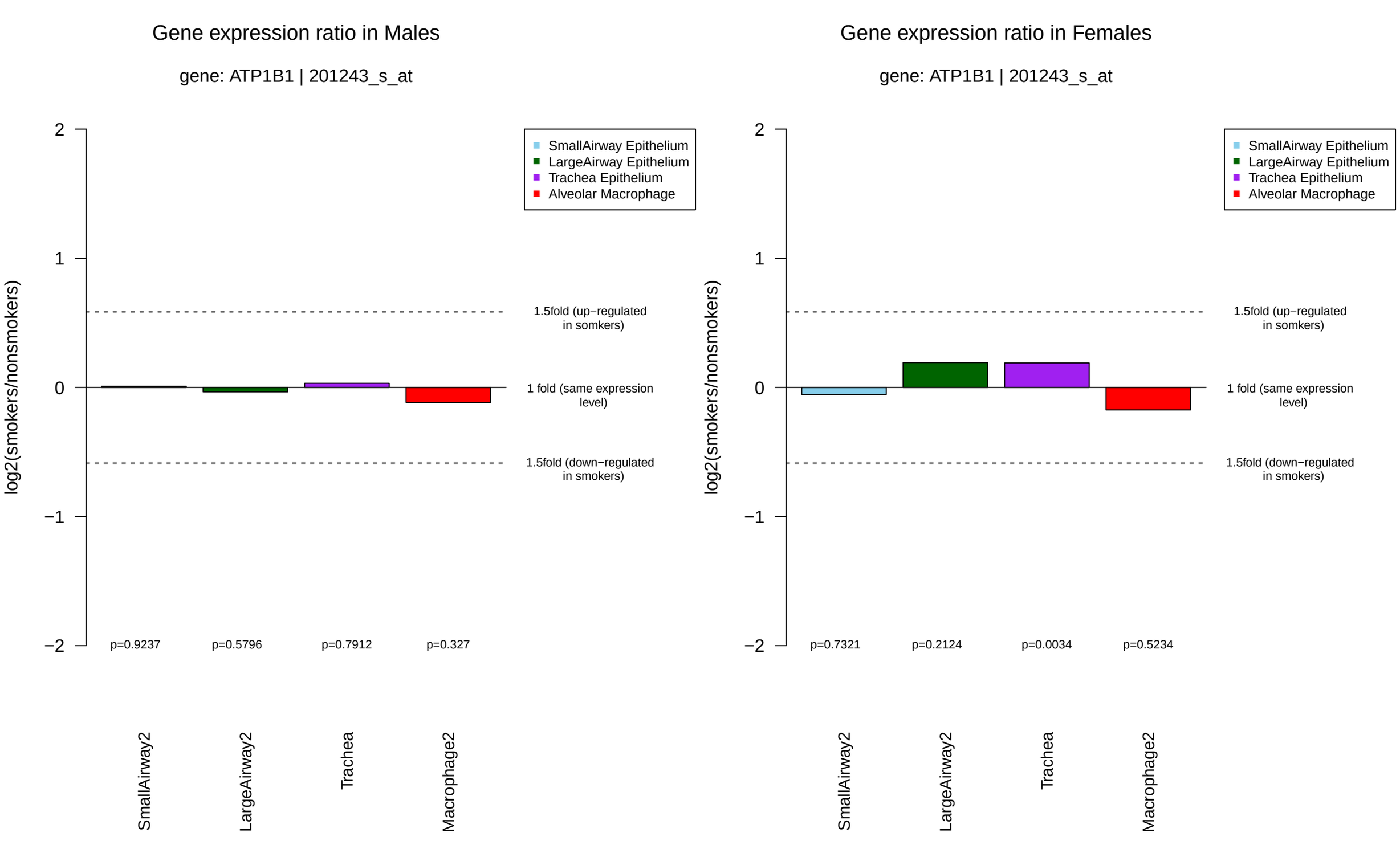
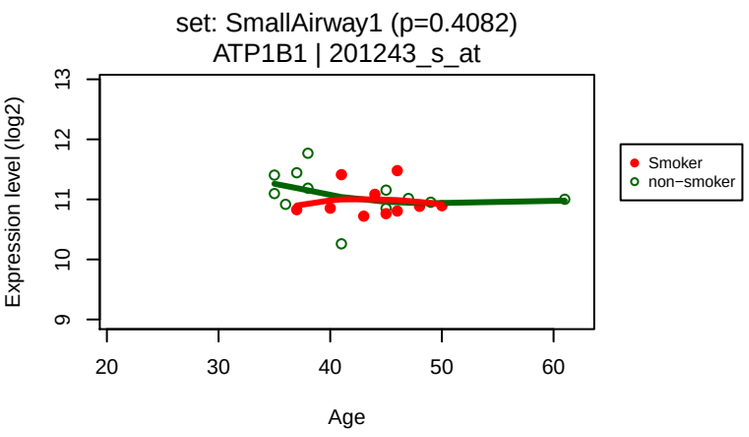
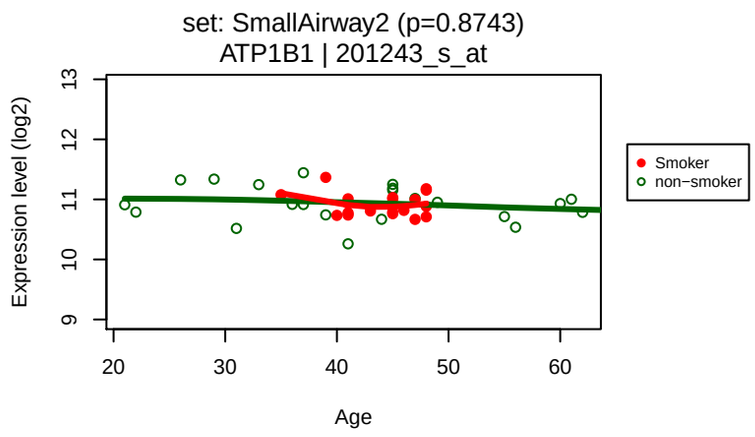
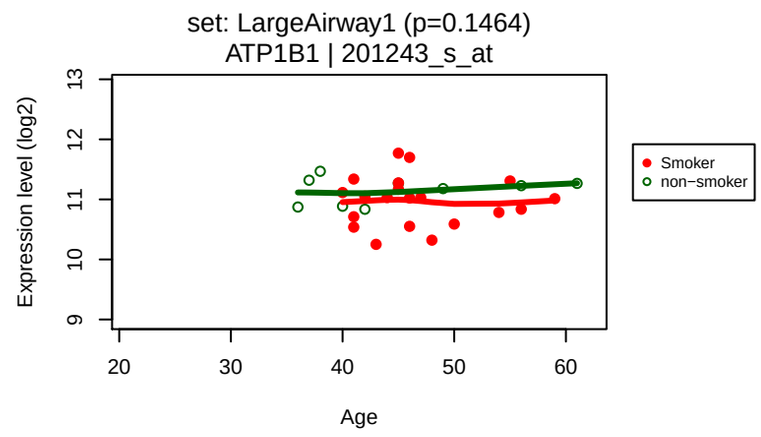

Gene expression of smokers and nonsmokers
gene: ATP1B1 | 201243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4082
p=0.8743
p=0.8612
p=0.1464
p=0.6059
p=0.5033
p=0.4
p=0.2063
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

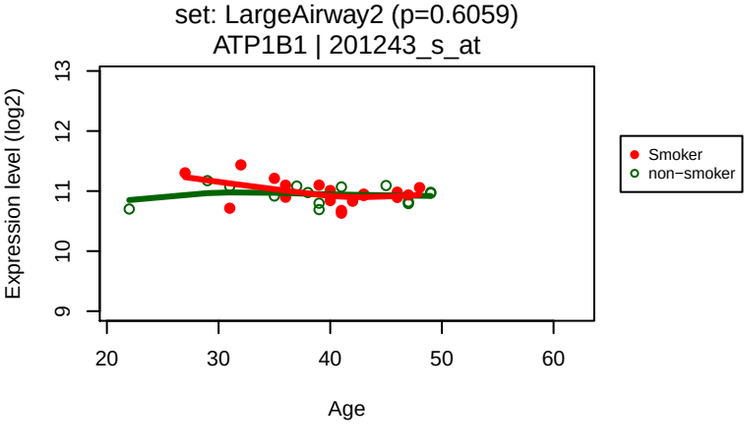

Gene expression ratio (smokers/nonsmokers)
gene: ATP1B1 | 201243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4082
p=0.8743
p=0.8612
p=0.1464
p=0.6059
p=0.5033
p=0.4
p=0.2063
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

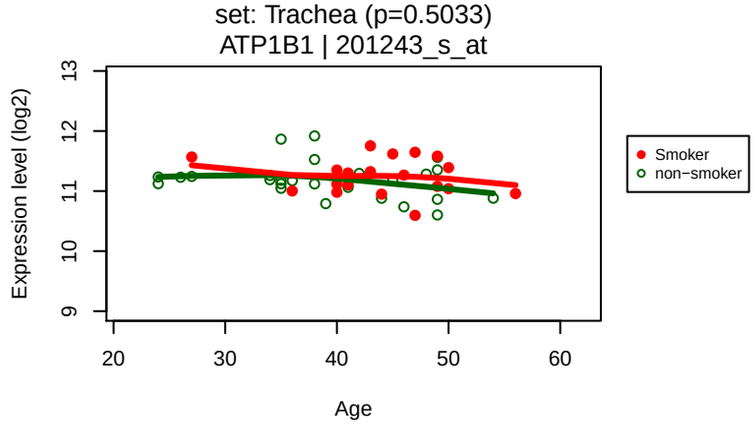

Gene expression ratio in Males
gene: ATP1B1 | 201243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9237
p=0.5796
p=0.7912
p=0.327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP1B1 | 201243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7321
p=0.2124
p=0.0034
p=0.5234
SmallAirway2
LargeAirway2
Trachea
Macrophage2

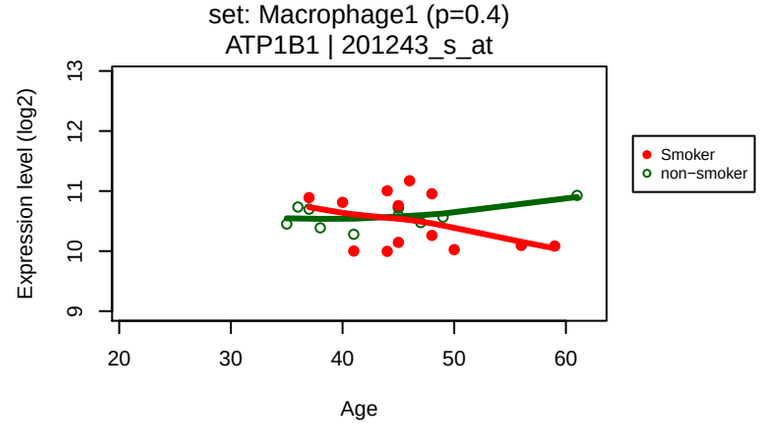

set: SmallAirway1 (p=0.4082)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker

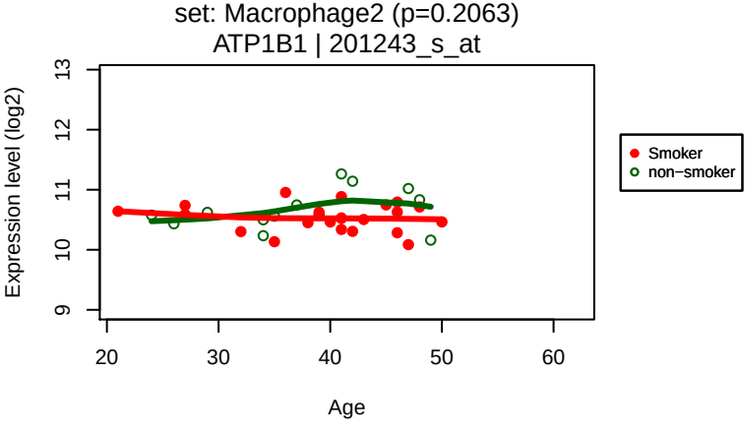

set: SmallAirway2 (p=0.8743)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker

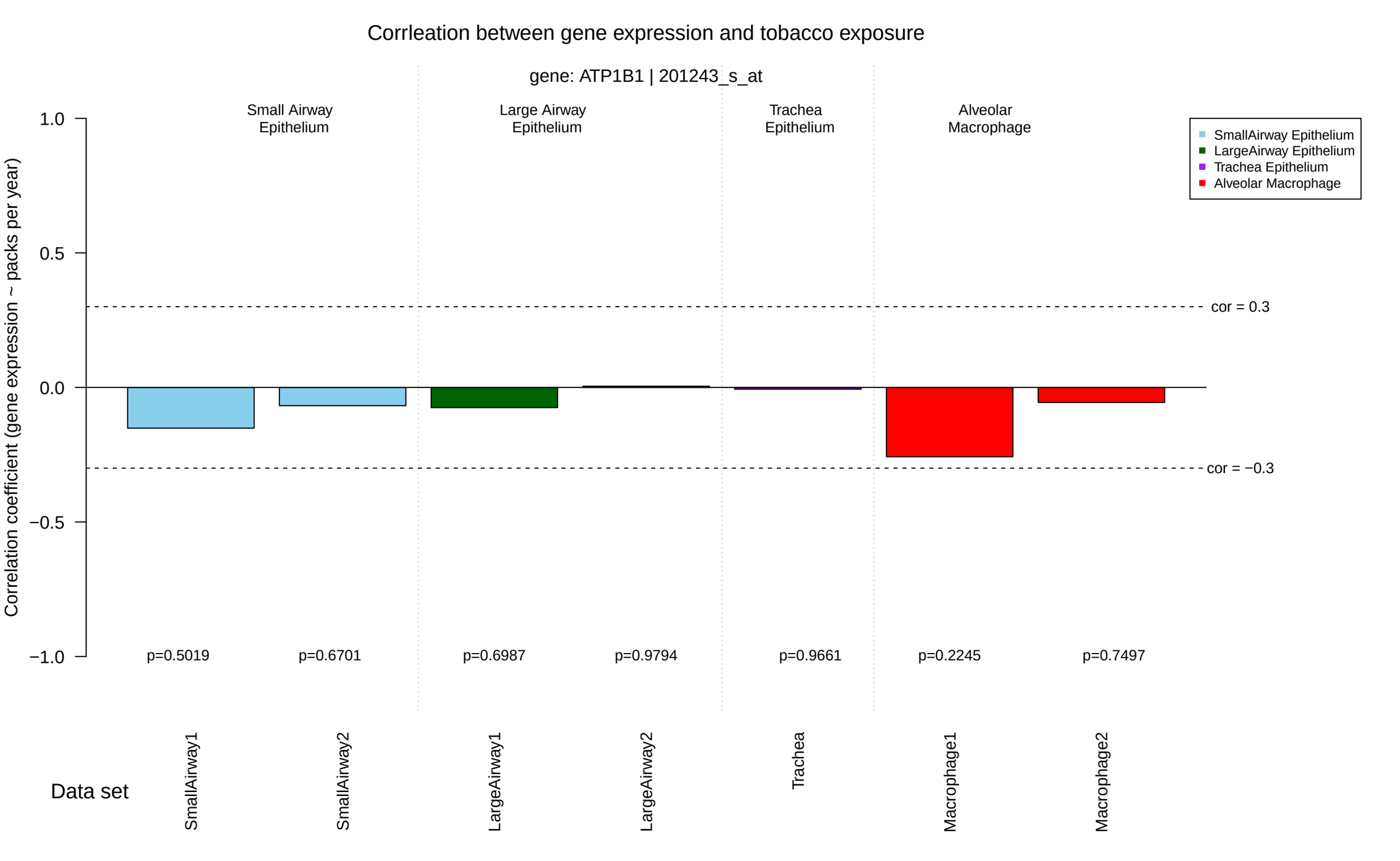

set: LargeAirway1 (p=0.1464)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6059)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5033)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2063)
ATP1B1 | 201243_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1B1 | 201243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5019
p=0.6701
p=0.6987
p=0.9794
p=0.9661
p=0.2245
p=0.7497
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2